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SUPERSEDING
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MILITARY SPECIFICATION

BEEF PATTIES, PRECOOKED, DEHYDRATED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers dehydrated precooked beef patties in flexible bags for use as a component of the meal, ready-to-eat, individual.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
EE-S-631	-	Spices, Ground and Whole and Spice Blends
QQ-A-1876	-	Aluminum Foil
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-26	-	Bags, Plastic, Polyethylene
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-C-1752	-	Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8905

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

THE U.S. PHARMACOPOEIAL CONVENTION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the U.S. Pharmacopoeial Convention, Inc., 46 Park Avenue, New York, NY 10016.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 882 - Tensile Properties of Thin Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 No. 19th St., Arlington, VA 22209.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 251 Kearney St. No. 410, San Francisco, CA 94108.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

* 3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and any of the following recognizable primal and sub-primal cuts: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, knuckles, and bottom rounds. Recognizable cuts are those which, when compared to IMPS cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. All suitable lean meat shall be used

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except tenderloins, and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which have been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed meat shall meet the limitations specified in tables I and II.

3.2.1.2 Grinding. The boned and trimmed beef shall be coarse ground through plates 1/2 to 1 inch in diameter. The analytical fat content of the coarse ground beef shall be not more than 23 percent nor less than 14 percent. Grinding shall be a continuous process.

* 3.2.1.3 Handling and storage of raw meat. Handling and storage of raw meat prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef shall be maintained in fresh chilled state at all time prior to boning and trimming.
b. Time between boning and coarse grinding shall not exceed 20 hours.
c. Beef processed through the cooking operation on the same day as boning and trimming shall not be allowed to exceed 50°F during processing.
d. Holding in the fresh chilled state for up to 4 days is permitted provided that the meat is maintained at a temperature not to exceed 40°F at all times prior to cooking.
e. Holding of coarse ground beef in the frozen state is permitted provided that:
   - beef is frozen to 0°F or lower within 72 hours after coarse grinding.
   - time between freezing of coarse ground beef and use in preparation of patties does not exceed 30 days.

3.2.2 Corn meal. The corn meal shall be a high commercial grade. Food grade pregelatinized corn meal (corn flour) may be used as an alternative. The granulation shall be such that not more than 7 percent is retained on a U.S. Standard No. 6 sieve. When mixed with water in the proportions specified in 3.3.1, the corn meal shall form a smooth paste indicating complete gelatinization.

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3.2.3 Onions, dehydrated. The onions shall be dehydrated minced onion or onion powder "fancy grade" of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association (ADOGA) for Dehydrated Onion and Garlic Products.

3.2.4 Salt. Salt shall be noniodized white, refined sodium chloride with or without anti-caking agents.

3.2.5 Pepper. The pepper shall be ground black pepper complying with type II of EE-S-631.

* 3.2.6 Water. Water used for ice making, formulation, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing.

* 3.3.1 Patty preparation. The patties shall be prepared as follows:

Formulation	Percent
Beef, coarse ground	91.00
Water	6.00
Corn meal	1.44
Salt	1.00
Black pepper	0.06
Dehydrated minced onion or onion powder	0.50

All of the ingredients except the beef shall be blended. If pregelatinized corn meal is not used, the blended ingredients shall be cooked until the starch in the corn meal is completely and smoothly gelatinized. The gelatinized corn meal blend shall then be evenly distributed with the coarse ground meat in equipment and then ground through a plate having holes 3/16 inch in diameter. The meat shall be continuously ground. Mechanical mixing may be used provided that the blending action does not result in fatty encapsulation of the meat which would affect its rehydration capability in the finished product. The ground mixture shall then be formed into rectangular patties of sufficient size that after cooking, the patties will conform to the dimension and weight requirements of 3.4.1. The patties shall be cooked immediately after forming.

3.3.2 Patty cooking. The formed patties shall be broiled to an internal temperature of 150°F minimum. The broiler shall be adjusted to produce a patty complying with the finished product requirements.

3.3.3 Patty freezing. The patties shall be placed in a blast freezer or individually quick frozen within 4 hours after completion of cooking and frozen to 0°F or lower within 6 hours after placement in the freezer. The temperature of the frozen patties shall not be allowed to exceed 5°F at anytime during handling and storage prior to dehydration.

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3.3.4 Patty handling conditions prior to dehydration. The time from cooking to start of dehydration shall not exceed 10 days.

3.3.5 Freeze dehydration. The patties shall be freeze-dehydrated at an absolute pressure not to exceed 1.5 millimeters of mercury and a patty temperature, as indicated by suitable instruments, of no more than 150°F. If the platen temperature is maintained at 155°F or below, the patty temperature may be disregarded. After dehydration has been completed, the pressure shall be equalized to atmospheric level with nitrogen and the patties shall be unit packed without delay as specified in 5.1. In no case shall more than 16 hours elapse between the time the chamber is opened and the time the patties are completely unit packed. During the interim, the patties shall be adequately protected from oxygen and moisture by either holding under a nitrogen atmosphere with 2.0 percent or less oxygen, or under a vacuum of at least 27 inches for the entire period. If vacuum is used, it shall be broken with nitrogen.

3.4 Finished product. The finished product shall comply with the requirements of 3.4.1 and 3.4.2.

3.4.1 Dehydrated product.

a. There shall be no foreign material, such as, but not limited to dirt, insect, insect part, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no foreign color, such as, but not limited to purple, blue, or green.

d. There shall be no evidence of incomplete dehydration such as damp or soggy areas.

e. There shall be no evidence of faulty dehydration such as glazed surface areas measuring more than 0.5 inch in any dimension or dark colored cores of any dimension.

f. Product shall have a characteristic grilled appearance. There shall be no burned areas evidenced by black color larger than 0.5 inch in any dimension. (Black color which disappears after the product has been rehydrated as indicated below, shall not be considered burnt.)

g. The product shall consist of patties meeting the following dimensions:

Length	Width	Thickness
2.5 to 4.25 inches	2.0 to 2.5 inches	0.5 to 0.9 inch

h. The dehydrated patty average weight shall be 34 grams minimum and no individual dehydrated patty shall weigh less than 28 grams.

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i. Moisture content of the dehydrated product shall not exceed 2.0 percent.

j. The bacterial content shall not exceed a standard plate count of 75,000 per gram.

k. The E. coli count shall be less than 3 per gram in at least four out of five samples and shall not exceed 10 per gram in any sample.

3.4.2 Rehydrated product.

a. When rehydrated in an excess of hot water (180°-120°F) for a maximum of 5 minutes, the product shall rehydrate completely, except for any crust caused by cooking.

b. The product shall be removable from the rehydration vessel in no more than four distinct pieces using a spatula or similar instrument and reasonable care.

c. The product shall contain no bone measuring more than 0.3 inch in any dimension.

3.4.3 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.5 Plant qualification. The meat component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. When a first article is required (see 6.1) it shall be inspected in accordance with the quality assurance provisions of this document, and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the AMS agent by the contractor. However, the AMS agent reserves the right to declare as a lot any portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or presence of any off-condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

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TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/ 3/
Major	Minor	
151		Presence of major lymph gland measuring 1.0 inch or more in any dimension.
	201	Presence of major lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
152		Presence of blood clot measuring 1.0 inch or more in any dimension.
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
153		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
154		Presence of bone measuring 0.3 inch or more in any dimension.
155		Presence of cartilage measuring 0.5 inch or more in any dimension.

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TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/ 3/
Major	Minor	
156		Presence of backstrap or other ligament measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap or other ligament measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
157		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank or abdominal section of short plate measuring 3 square inches or more.
158		Presence of knuckle cover.
159		Presence of kidney, pizzle eye, pre-pubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, pre-pubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in longest dimension.

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TABLE II. Boned and trimmed beef defects (cont'd)

Item No.		Defect 1/ 2/ 3/
Major	Minor	
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.
	213	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	214	Presence of exposed major blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney or pelvic fat.

1/ Acceptability of product with respect to the presence of foreign material (for example: glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection employee. The presence of any product not in excellent condition shall cause rejection of the lot.

2/ Evidence of freezing or defrosting, or presence of any off-condition (including but not restricted to being off-odor, slightly sticky, gassy, rancid, sour, or showing discoloration) shall cause rejection of the lot.

3/ This table contains a listing of standard defects some of which may not apply to product which has been boned and trimmed from the specific cuts specified in 3.2.1. The listing of such defects does not relieve the contractor of the responsibility of complying with the requirements of 3.2.1.

4.5.1.3 Other ingredients examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

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4.5.1.4 Laminated bag material certification. Material listed below shall be accepted on the basis of a contractor's certificate of conformance to the indicated requirements.

Material requirement	Requirement paragraph	Test procedures
Ionomer or polyethylene film thickness	5.1.1	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Polyester film thickness	5.1.1	As above.
Aluminum foil thickness	5.1.1	As specified in QQ-A-1876
Laminated material construction	5.1.1	Laboratory evaluation
Color of laminated material	5.1.1	Visual evaluation

* 4.5.1.5 Polyethylene bag examination. The unfilled bags shall be examined for the defects listed in table III. The lot size shall be expressed in bags. The sample unit shall be one unfilled bag. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.65.

TABLE III. Unfilled polyethylene bag defects

Category	Defect
Major	
151	Not size specified
152	Individual bags that stick together or that tear when specified
153	Not material specified
154	Tear, hole, or open seal
155	Not clean

* 4.5.1.6 Preformed laminated bag examination (when applicable). The unfilled bags shall be examined for the defects listed in table IV. The lot size shall be expressed in bags. The sample unit shall be one unfilled bag. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 0.65. For major defects and 4.0 for minor defects.

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TABLE IV. Unfilled preformed laminated bag defects

Category		Defects
Major	Minor	
151		Not size specified
152		Heat seal dimensions on three edges not as specified
153		Tear, hole, or open seal
154		Not clean
155		Color not as specified
156		Required labeling missing, incorrect, or illegible
157		Not material specified
158		Individual bags that stick together or that tear when separated
	201	Tear notch or serrations missing
	202	Tear notch or serrations not located as specified
	203	Depth of tear notch or serrations not as specified

4.5.1.7 Preformed laminated bag seal strength testing (when applicable). The unfilled bags shall be tested for seal strength in accordance with Method A or B of ASTM D 882 except that testing speed may be 10 to 12 inches per minute. The lot size shall be expressed in bags. The sample size shall be the number of bags indicated by inspection level S-1. Three adjacent specimens shall be cut from each of the three sealed sides of each bag in the sample. The results shall be reported to the nearest 0.1 pound. The average seal strength of each seal shall be calculated by averaging the strengths of the three test specimens cut from that seal. Any test specimen failing to meet the individual test specimen seal strength requirement specified in 5.1.1 or any seal failing to meet the average seal strength requirement specified in 5.1.1 shall be cause for rejection of the lot.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to preparation, processing, holding, bag filling, and bag closing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

* 4.5.3 In-process testing for fat content. The raw coarse ground beef shall be tested for fat content in accordance with the crude fat method (using petroleum ether) in the meat section of the Meat and Meat Products chapter in the Official Methods of Anlysis of the Association of Official Analytical Chemists. Failure of one or more of the four samples shall be cause for rejection of the raw beef involved. Fat results shall be reported to the nearest 0.1 percent.

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4.5.4 Patty weight inspection. The weight of the dehydrated patties shall be determined by weighing each sample unit on a suitable scale tared with an average weight of representative empty laminated bags and polyethylene bags. Any individual weight of less than 28 grams shall be scored as a minor defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. In addition, the lot shall be rejected if the sample average patty weight is less than 34 grams.

* 4.5.5 Laminated bag vacuum examination. The filled and sealed bags shall be visually examined for a proper vacuum level not less than 48 hours after filling and sealing. The sealed bag shall continue to exhibit tight adherence to the surface contours of the contents when a pulling force is applied at the center of each side seal. This force shall be applied by holding each side seal between thumb and forefinger of each hand, simultaneously exerting a slight pull with both hands. Any evidence of loss of vacuum shall be considered a major defect. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I (but not less than 200 sample units) and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

4.5.6 Filled and sealed laminated and polyethylene bag examination. The filled and sealed laminated and polyethylene bags shall be examined for the defects listed in table V. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be I (but not less than 200 sample units) and the AQL, expressed in terms of defects per hundred units, shall be 0.65 for major defects and 2.5 for minor defects.

TABLE V. Filled and sealed laminated and polyethylene bag defects 1/

Category		Defect
Major	Minor	Laminated bag
151		Not formed as specified 2/
152		Not clean 2/
153		Tear, hole, or open seal 2/
154		Closure seal not produced by specified method
155		Evidence of delamination 3/
156		Closure seal width not as specified
157		Closure or top seal extends into or below tear notch 4/
158		Bag has foreign odor
159		Required labeling missing, incorrect, illegible, or that smudges
160		Color not as specified 2/
161		Tear notch or serrations missing
162		Tear notch or serrations not located as specified

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TABLE V. Filled and sealed laminated and polyethylene bag defects 1/ (cont'd)

Category		Defect
Major	Minor	Laminated bag (cont'd)
163		Depth of tear notch or serrations not as specified
164		Inner bag caught in closure seal making removal of contents impossible without rendering the laminated bag unfit for rehydration of the contents
165		Presence of flex fractures in the aluminum foil 2/ 5/
	201	Inner bag caught in closure seal, but contents can be removed without making laminated bag unfit for rehydration of the contents
		Polyethylene bag
	202	Open end of bag not closed by folding end back over the patty

1/ Any evidence of insect or rodent infestation shall be cause for rejection of the lot.

2/ Applicable to form-fill-seal bags only.

3/ Delamination shall be scored as a defect except delamination of outer ply when located in the seal area 1/16 inch or further from food product edge of seal. Bags exhibiting this type of delamination shall be tested by manually flexing the delaminated area 10 times. The area of delamination shall be held between the thumb and forefinger of each hand with both thumbs and forefingers touching each other. The delaminated area shall then be rapidly flexed by rotating both hands in alternating clockwise - counterclockwise directions. Care shall be exercised when flexing delaminated area near the tear notches to avoid tearing the bag material. After flexing, the separated outer ply shall be grasped between the thumb and forefinger and gently lifted toward the food product edge of the seal. If the separated area is too small to be held between thumb and forefinger, a number two stylus shall be inserted into the delaminated area and a gentle lifting force applied against the outer ply. If separation of the outer ply can be made to extend to less than 1/16 inch from the product edge of the seal with no discernible resistance to the gentle lifting, the bag shall be rejected.

4/ Applicable to preformed bags only.

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5/ To test for flex fractures the inside surface of the tray-shaped body shall be placed over a light source and the outside surface observed for the passage of light. Observation of light through the bag material shall be evidence of the presence of flex fractures in the aluminum foil.

4.5.7 Product examination. The finished product shall be examined for the defects listed in table VI. The lot size shall be expressed in bags. The sample unit shall be the contents of one filled and sealed bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE VI. Product examination 1/ 2/

Category		Defect
Major	Minor	Applicable to dehydrated product:
151		Evidence of incomplete dehydration such as damp or soggy areas.
152		Presence of glazed surface areas measuring more than 0.5 inch in any dimension.
153		Presence of burned areas as indicated by a black spot greater than 0.5 inch in any dimension (see 3.4.1).
154		Presence of dark colored cores of any dimension.
155		Crushed 3/
	201	Length of patty less than 2.5 inches or more than 4.25 inches.
	202	Width of patty less than 2.0 inches or more than 2.5 inches.
	203	Thickness of patty less than 0.5 inch or more than 0.9 inch.
		Applicable to rehydrated product 4/:
156		Not completely rehydrated within a maximum period of 5 minutes.

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TABLE VI. Product examination 1/ 2/ (cont'd)

Category		Defect
Major	Minor	Applicable to rehydrated product 4/: (cont'd)
157		Not removable from rehydration vessel without breaking into more than four distinct pieces.
158		Presence of bone measuring more than 0.3 inch in any dimension.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ A crushed patty is one in which 1/8 or more of the volume of the patty has been reduced to crumbs or powder.

4/ Rehydrated in an excess of hot water (180° to 210°F) for a maximum of 5 minutes.

* 4.5.8 Laminated bag closure seal testing. The filled and sealed bags shall be tested for closure seal strength in accordance with method A or B of ASTM D 882 except that the testing speed may be 10 or 12 inches per minute. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed bag. The inspection level shall be S-3 and the AOL, expressed in terms of defects per hundred units, shall be 1.5. For the preformed bag, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the top closure seam of each bag in the sample. For the alternate form-fill-seal bag, which has all four sides closed simultaneously, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from each of the four sides of each bag in the sample. The results shall be reported to the nearest 0.1 pound per linear inch. For the preformed bag the average seal strength of the closure seal shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form, fill, seal bag the average seal strength of each closure seal shall be calculated by separately averaging the test results of the three test specimens cut from each side seal. Any test specimen failing to meet the individual test specimen seal strength requirement specified in 5.1.1 or any closure seal failing to meet the average seal strength requirement specified in 5.1.1 shall be scored a defect.

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4.5.9 Moisture content testing. The finished product shall be tested for moisture content in accordance with the Moisture - Drying in Vacuum method in the Meat section of the Meat and Meat Products chapter in the Official Methods of Analysis of the Association of Official Analytical Chemists. A temperature of 70°C for 16 hours under a pressure of 100 mm of mercury shall be used. The results shall be reported to the nearest 0.1 percent. Any result not conforming to the requirement in 3.4.1 shall be considered a defective. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed laminated bag. The inspection level shall be S-2 and the AQL, expressed in terms of percent defective, shall be 1.5.

* 4.5.10 Microbiological testing. Five filled and sealed bags shall be selected at random from the lot regardless of lot size. The contents of each sample bag shall be tested in accordance with the Examination of Frozen, Chilled, Precooked, or Prepared Foods section in the Microbiological Methods chapter of the Official Methods of Analysis of the Association of Official Analytical Chemists except that each sample shall be prepared for testing by being adequately comminuted with care being taken to avoid contamination of product and moisture pickup. For E. coli determinations, proper dilutions shall be prepared using 25-gram samples. The diluent shall be added to the sample of dry product and allowed to stand for 15 minutes before blending. Any of the following results shall be considered a test failure:

a. Any of the five samples has a confirmed E. coli count of greater than 10 per gram.
b. Two or more of the five samples have a confirmed E. coli count equal to or greater than 3 per gram.
c. The total plate count of any sample exceeds 75,000 per gram.

Any test failure shall be cause for rejection of the lot.

4.5.11 Shipping container examination. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria contained in the appendix of PPP-B-636, with the following defects added to the table: Major: (1) Marking missing, incorrect, or illegible, (2) Cushioning material missing or not as specified; Minor: (1) Less than or more than 5 percent of specified number, (2) Bags not packed flat in layers.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. Each patty shall be placed in a (minimum) 0.003 inch thick, food grade, polyethylene bag fabricated in accordance with type II, style 1 of PPP-B-26. The polyethylene bag shall be of such size as to facilitate bag filling, allow for bag closure as specified, and facilitate subsequent filling and closure of the laminated bag with no obstruction from excess material. Each patty shall be

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placed into the polyethylene bag with one complete edge against the bag bottom. The open end of the bag shall be folded against the nearest patty edge and extend over one of the largest adjacent patty surface a minimum of 1-1/2 inches and a maximum of the whole length or width of that surface as applicable. The patty in the polyethylene bag shall be unit packed in a laminated bag. The bag material shall be a 3 ply laminate consisting of 0.002 inch thick ionomer or polyethylene film laminated or extrusion coated to 0.00035 inch thick aluminum foil which is then laminated to 0.0005 inch thick polyester. The three plies shall be laminated so that the aluminium foil is between the other two layers. The bag shall be formed with the polyester on the exterior of the bag. The complete exterior surface of the bag shall be uniformly colored in the range of green 34079 through 34087 or 24052 through 24087 or brown 30045 through 30118 (excluding 30109) or 10045 of FED-STD-595. The laminated bag shall be formed by heat sealing to the size and design construction as shown in figure 1. The laminated bag may be preformed (sealed on three sides) then filled and sealed at the open end, or, the laminated bag may be formed, filled and the seals simultaneously applied on all four sides. The end of the laminated preformed bag may be provided with a 1/4 $\pm$ 1/16 inch extended or fold-over lip to facilitate opening and filling. Tear notches as indicated in figure 1 shall be provided on preformed bags or bags that are formed, filled and simultaneously sealed on all four sides. Alternatively, if the bag has serrated edges, the serrations may be used as tear notches provided that the serrations are sharp (no plastic tailings exist) and the serration depth and the minimum seal width at the serrated edges are in accordance with the dimensions shown in figure 1. The filled bag shall be closed under a vacuum of not less than 25 inches of mercury (see 4.5.5). Closure of the preformed bag shall be effective with a continuous heat seal not less than 1/4 inch wide. If thermal impulse or combination (heated curved bar with thermal impulse) sealing is used, any seal width from 1/8 to 7/16 inch will be acceptable. The closure or top seal location shall be as shown in figure 1. Closure of the form fill seal bag shall be accomplished by application of all four seals simultaneously. All four seals shall be 3/8 $\pm$ 1/8 inch wide. The material shall show no evidence of delamination or degradation when heat sealed or fabricated into bags and shall not transfer any foreign odor or flavor to the unit packed product. The average seal strength of each seal shall be not less than 6 pounds per linear inch, and no individual test specimen shall be less than 5 pounds per linear inch when tested as specified in 4.5.1.7 and 4.5.8.

5.2 Packing. Packing shall be level C.

* 5.2.1 Level C, for shipment to ration assembly. Not more than 325 patties preserved as specified in 5.1.1 shall be packed flat in a fiberboard box conforming to style RSC-L, class domestic, variety SW, grade 175 of PPP-B-636. Cushioning material not less than 1/8 inch thick, conforming to type II, class B of PPP-C-843 or cushioning material not less than 1/16 inch thick conforming to PPP-C-1797 or cushioning material conforming to type 7, class 1 of PPP-C-1752, shall be placed between each of the layers and in the bottom and top of the box to minimize movement of the individual packages.

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5.3 Labeling and marking.

5.3.1 Unit packs. Labeling shall be in accordance with FDA and USDA labeling regulation. Each unit pack (bag) shall be clearly printed with permanent ink, in large clear black letters with the following information:

BEEF PATTY

EAT DRY OR RECONSTITUTE IN HOT OR COLD WATER. TO RECONSTITUTE, COVER PATTY WITH WATER. LET SOAK UNTIL SOFT (1-5 MINUTES). DRAIN.

Name and address of producer.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. Provisions for approved preproduction samples (see 3.4.3 and 6.3).

6.2 Award of contract. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions in 3.5.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 50

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DP - SS

Preparing activity:

Army - GL

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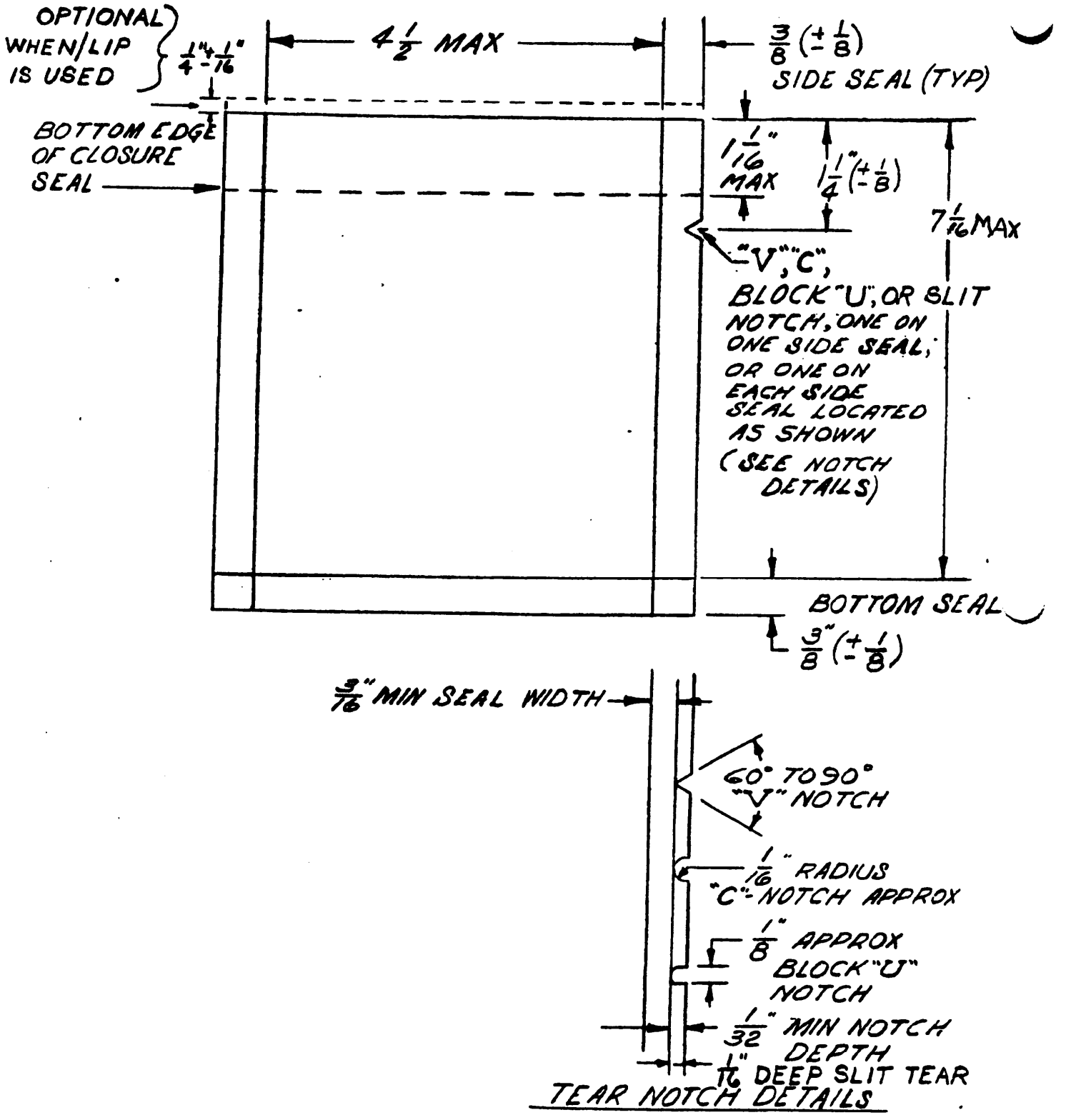


FIGURE 1
FOUR SEAL BAG

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44078A	Beef Patties, Precooked, Dehydrated, For Meal, ~~Ready-To-Eat~~

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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